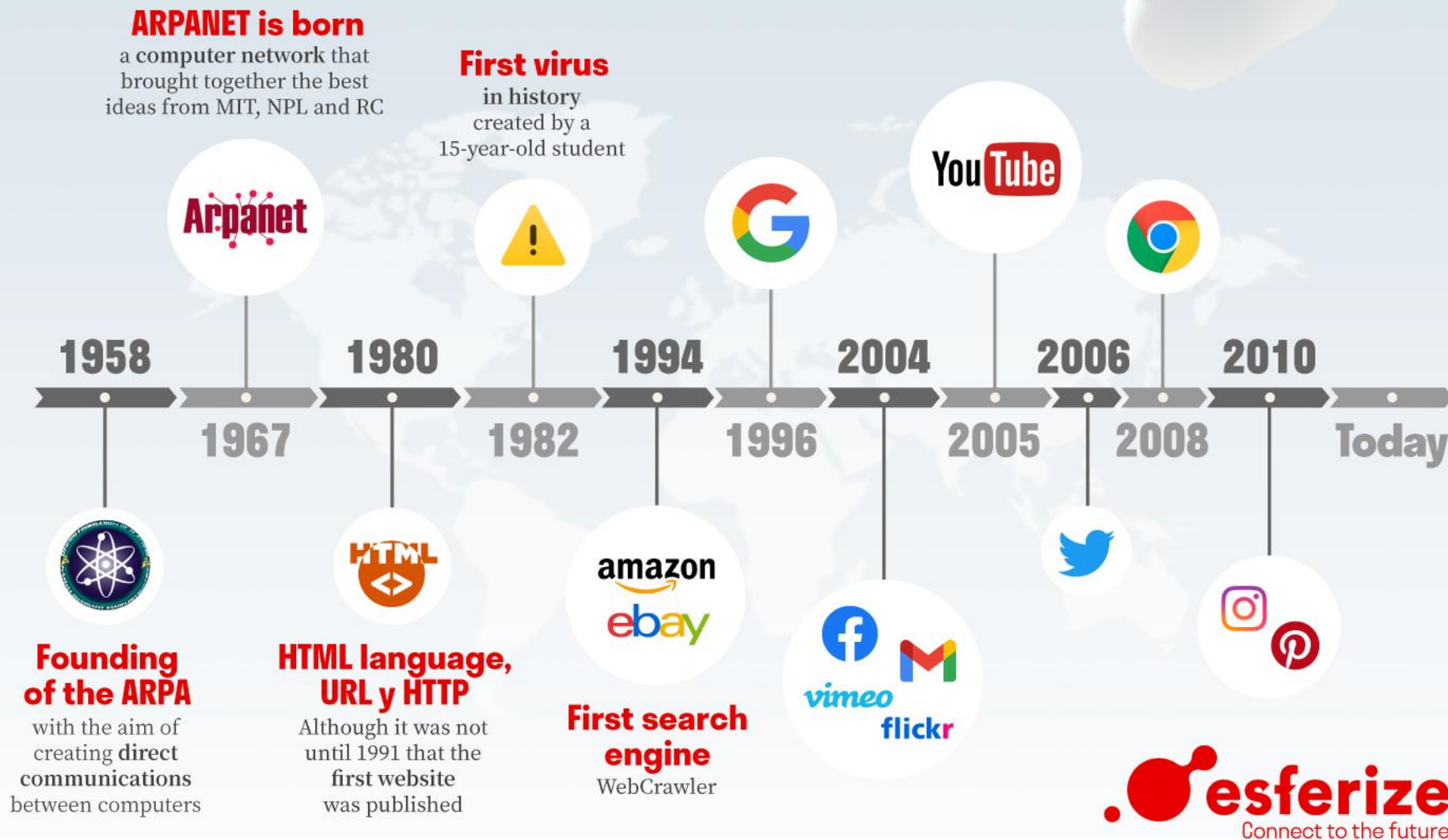
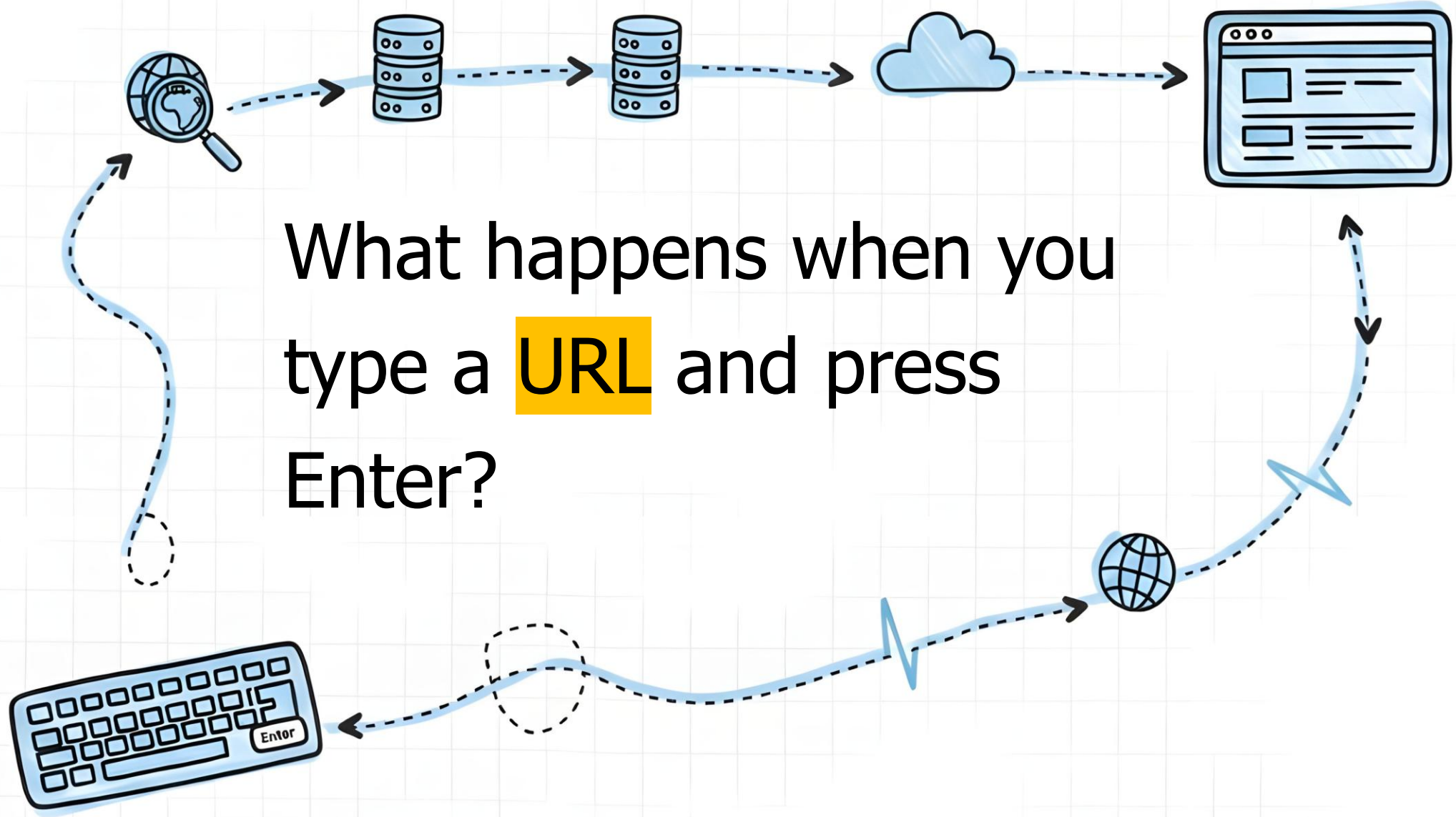


Web Technologies

Yet another background skill for the Digital Humanities

Evolution of the Internet

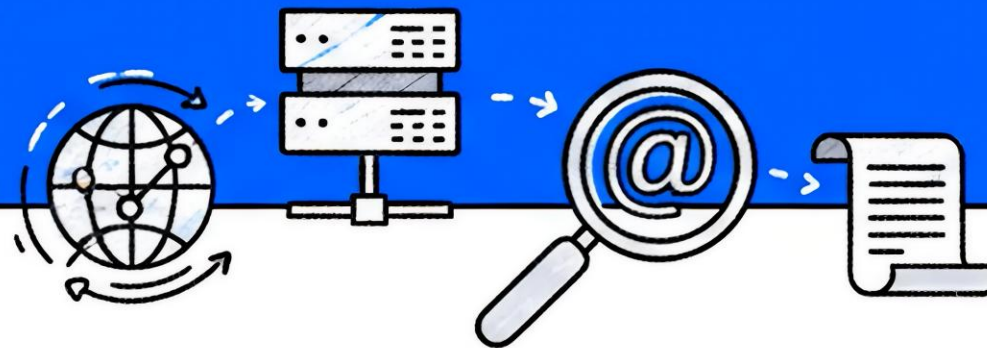




1

ENTER

A Click's Journey



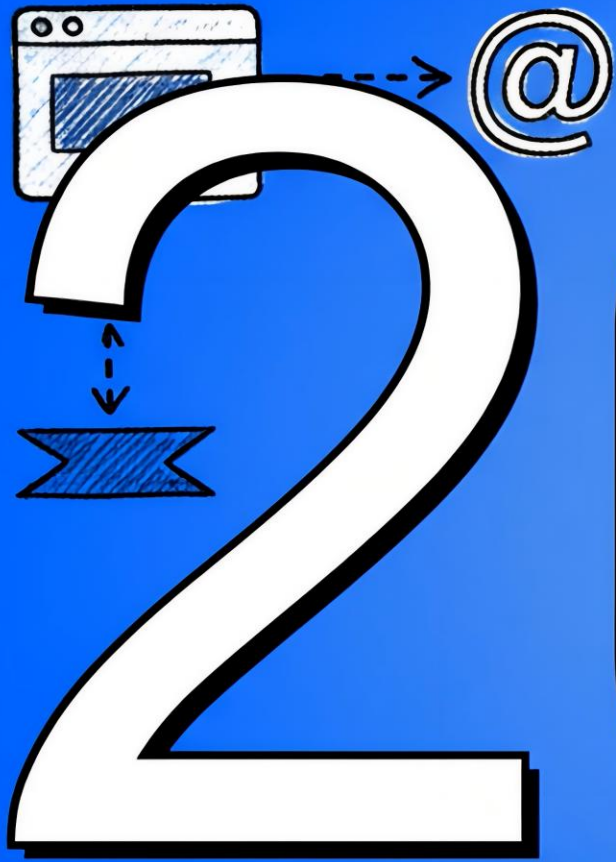
Anatomy of a URL

`https://example.com/path/to/something`

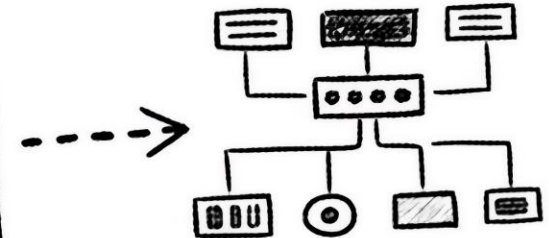
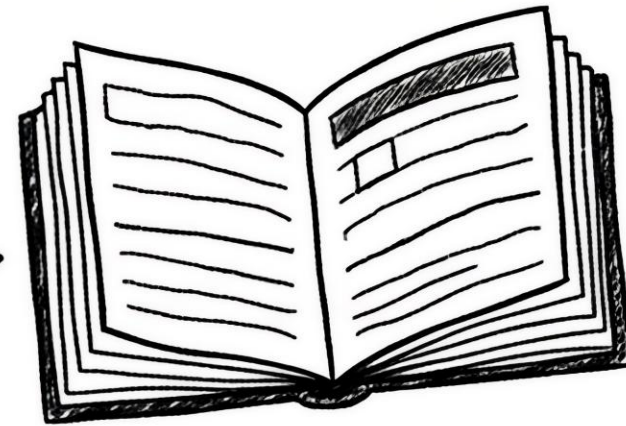
Protocol: `https://` - The protocol for secure communication

Domain: `example.com` – The human-friendly address of the website

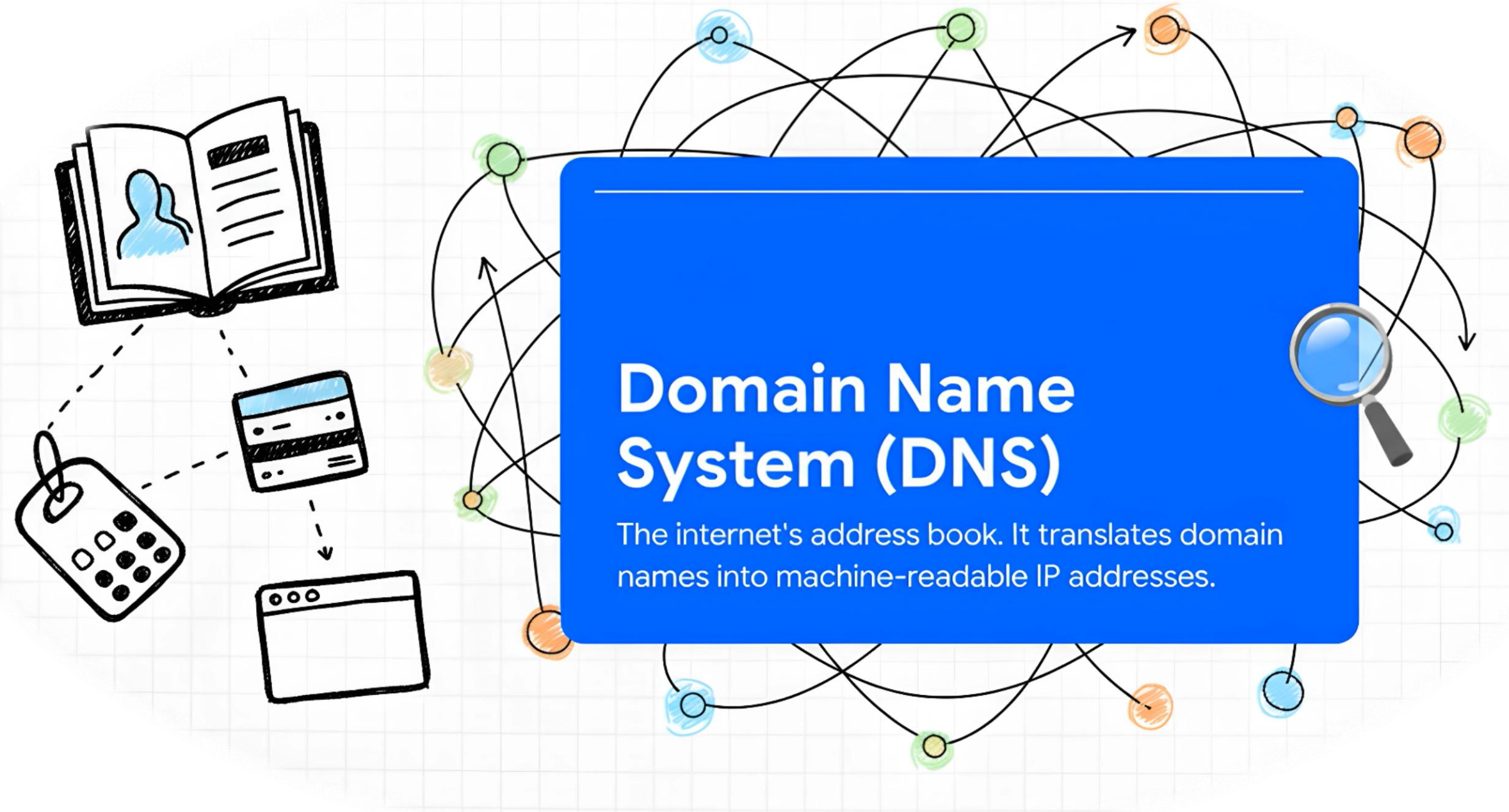
Path: `/path/to/something` – The specific resource or page you want to access on that site



The Internet's Address Book

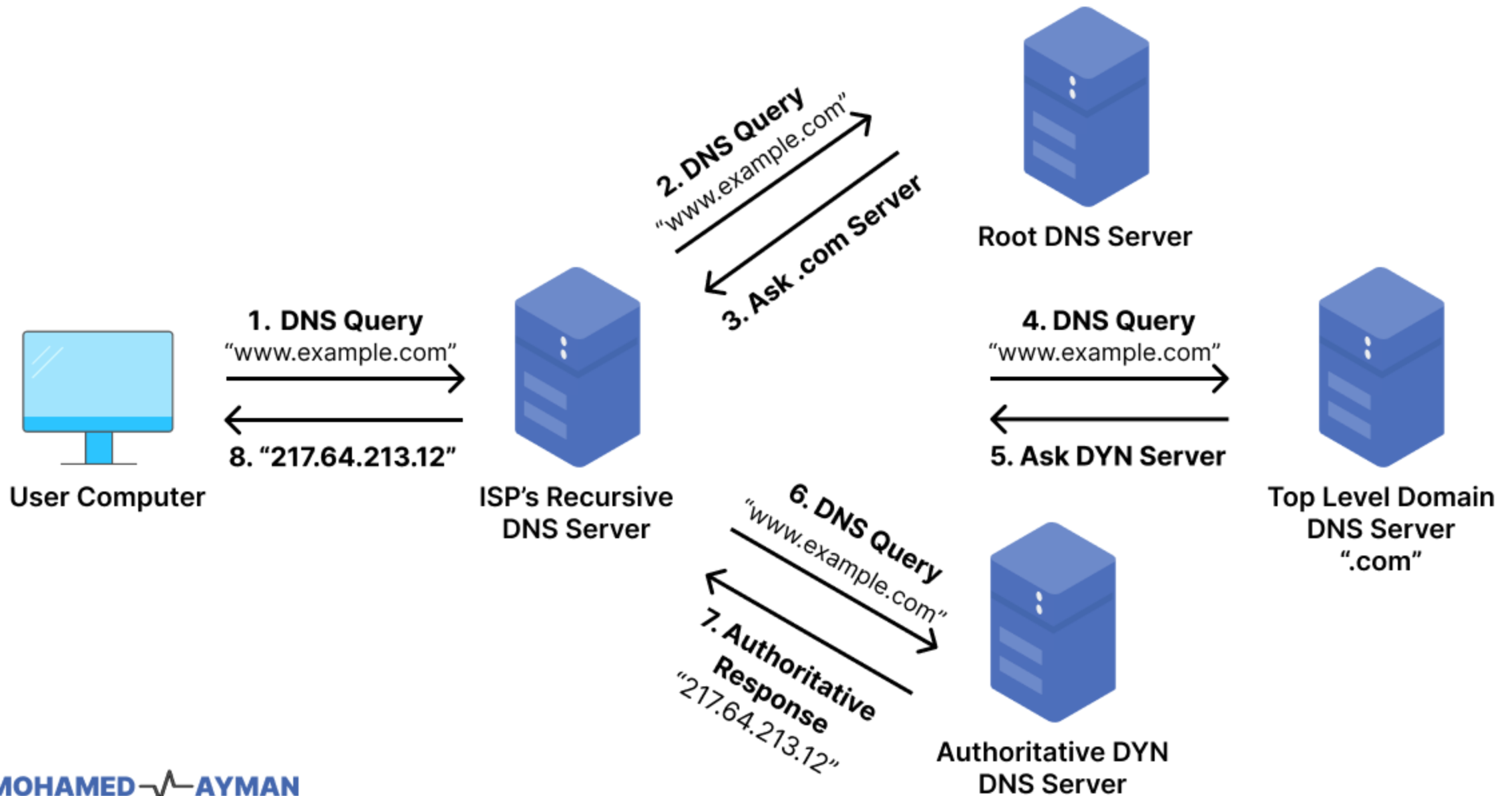


192.168.1.1



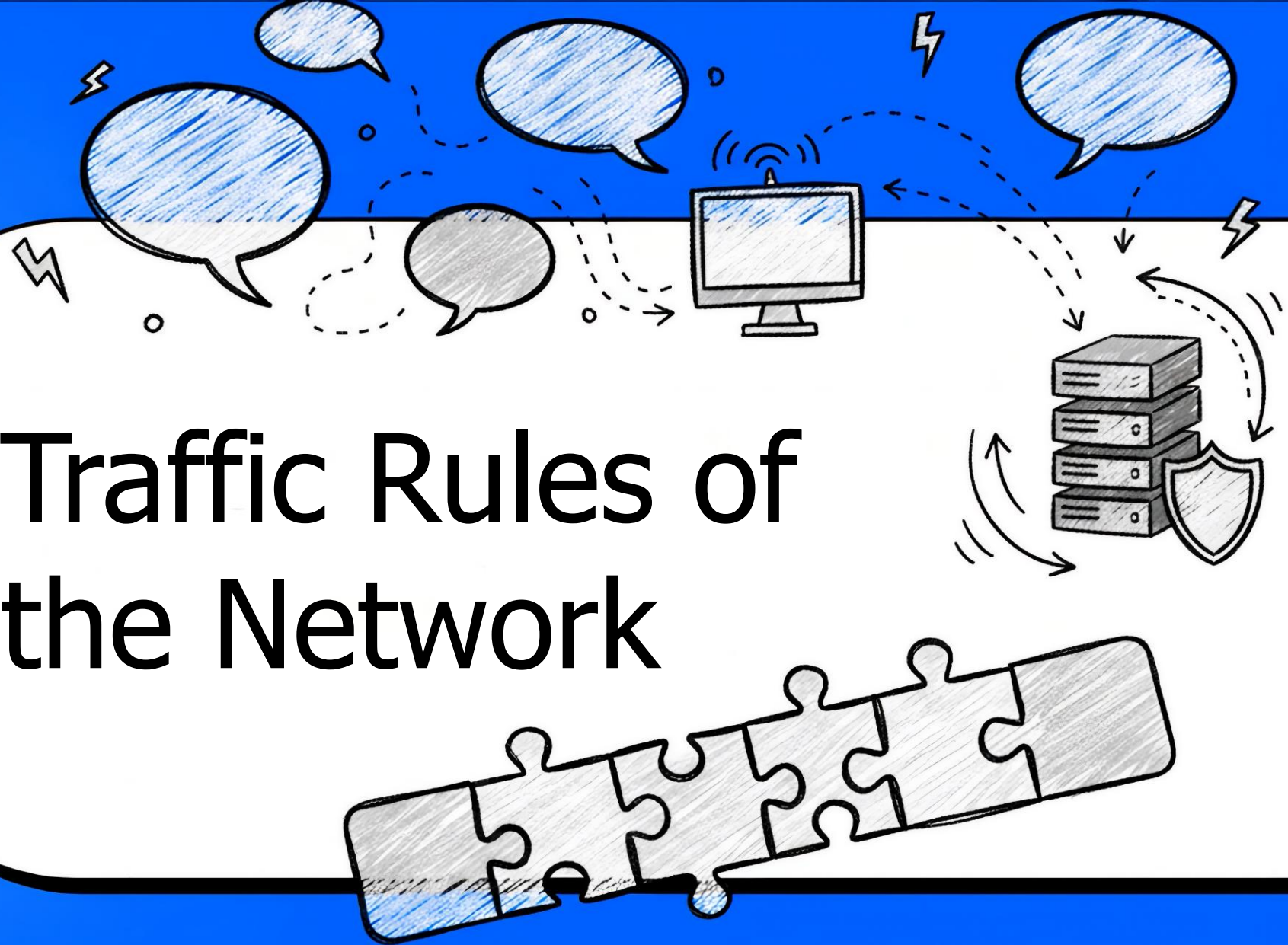
Domain Name System (DNS)

The internet's address book. It translates domain names into machine-readable IP addresses.



3

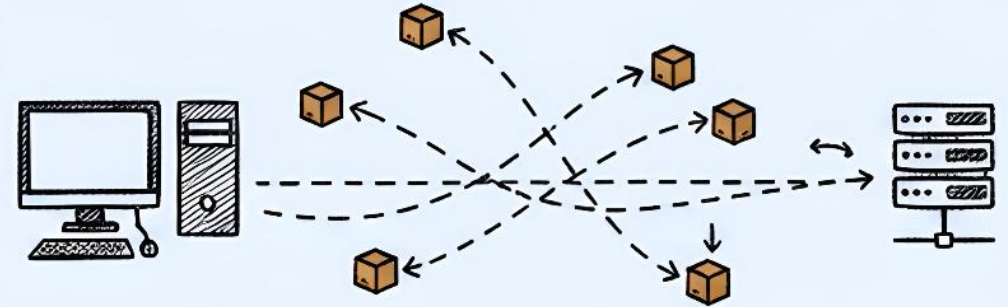
Traffic Rules of the Network



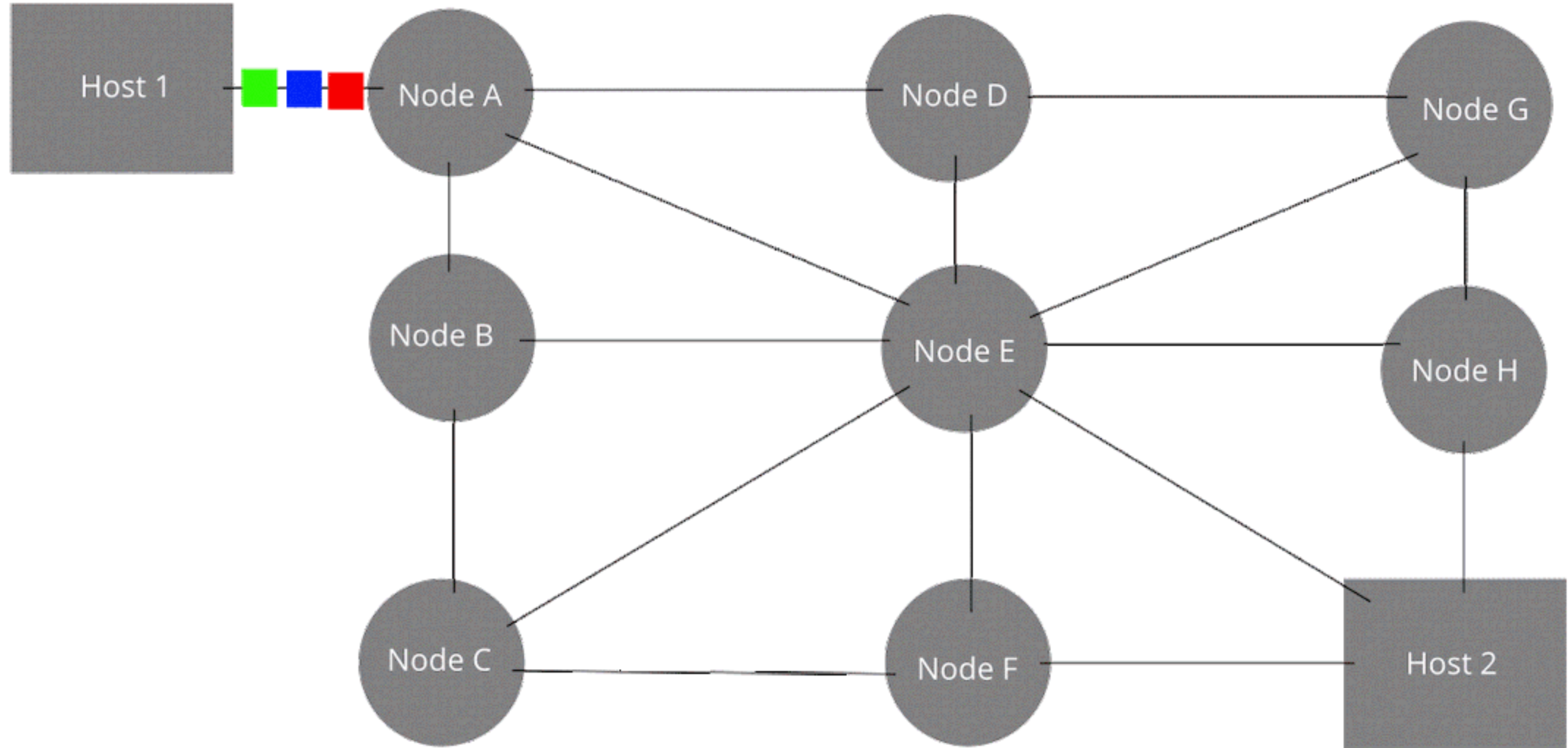
Circuit Switching



Packet Switching



The original message is Green, Blue, Red.





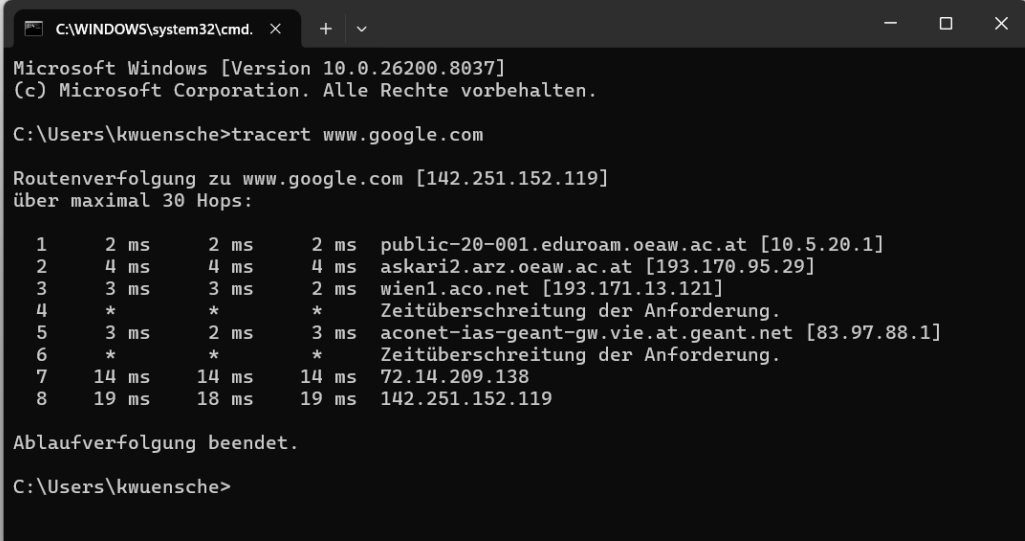
How to track your packets

tracert www.google.com (Windows) or

traceroute www.google.com (Mac/Linux)

You can visualize the
output here:

[https://stefansundin.github.io/
traceroute-mapper/](https://stefansundin.github.io/traceroute-mapper/)



```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. Alle Rechte vorbehalten.

C:\Users\kwwuensche>tracert www.google.com

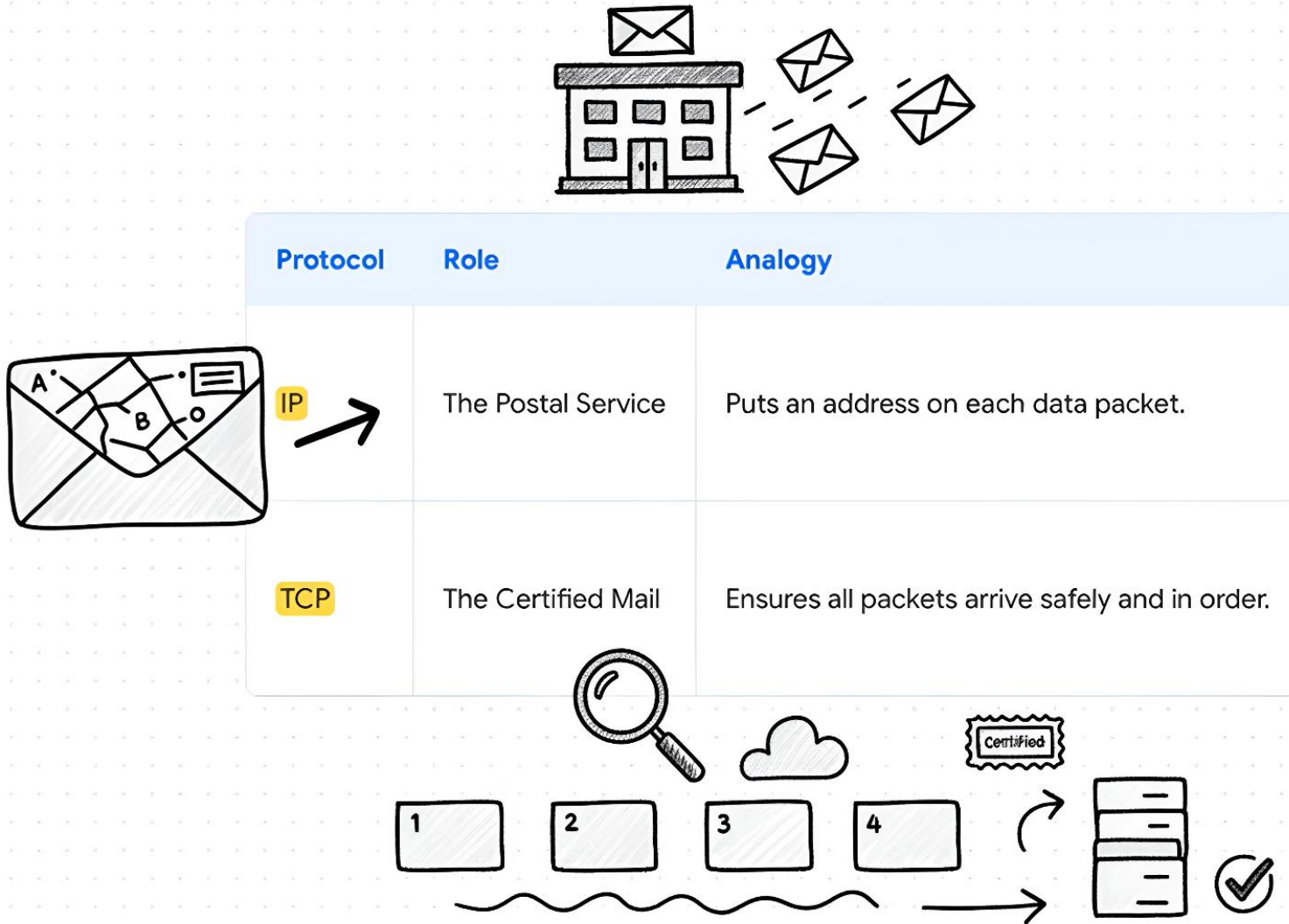
Routenverfolgung zu www.google.com [142.251.152.119]
über maximal 30 Hops:

 1    2 ms    2 ms    2 ms    public-20-001.eduroam.oeaw.ac.at [10.5.20.1]
 2    4 ms    4 ms    4 ms    askari2.arz.oeaw.ac.at [193.170.95.29]
 3    3 ms    3 ms    2 ms    wien1.aco.net [193.171.13.121]
 4    *      *      *      Zeitüberschreitung der Anforderung.
 5    3 ms    2 ms    3 ms    aconet-ias-geant-gw.vie.at.geant.net [83.97.88.1]
 6    *      *      *      Zeitüberschreitung der Anforderung.
 7   14 ms   14 ms   14 ms    72.14.209.138
 8   19 ms   18 ms   19 ms    142.251.152.119

Ablaufverfolgung beendet.

C:\Users\kwwuensche>
```


How does this work? TCP/IP!



IP Addresses

IPv4

192.168.0.1

4 blocks with 8 bits each

$\sim 4.3 \times 10^9$ possible addresses

IPv6

2001:0db8:85a3:08d3:1319:8a2e:0370:7344

8 blocks with 16 bits each

$\sim 3.4 \times 10^{38}$ possible addresses



4

Around the
World

The Physical Internet

Modem & Router: Your gateway to the internet.

Fiber-Optic Cables: Data travels as light through undersea cables.

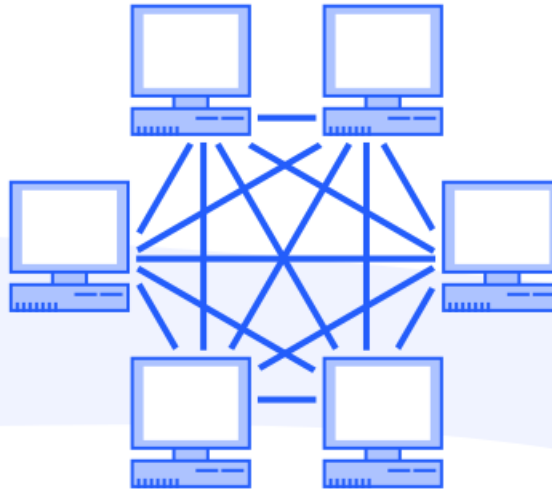
Switches & Routers: Devices that direct traffic between networks.

Internet Exchange Points: Hubs where different networks connect.

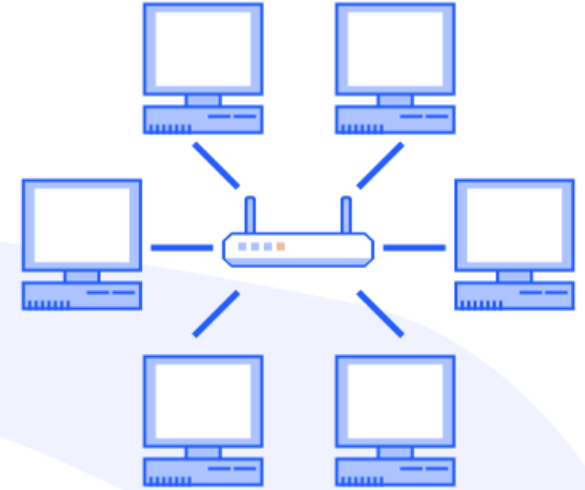
A Network of Networks



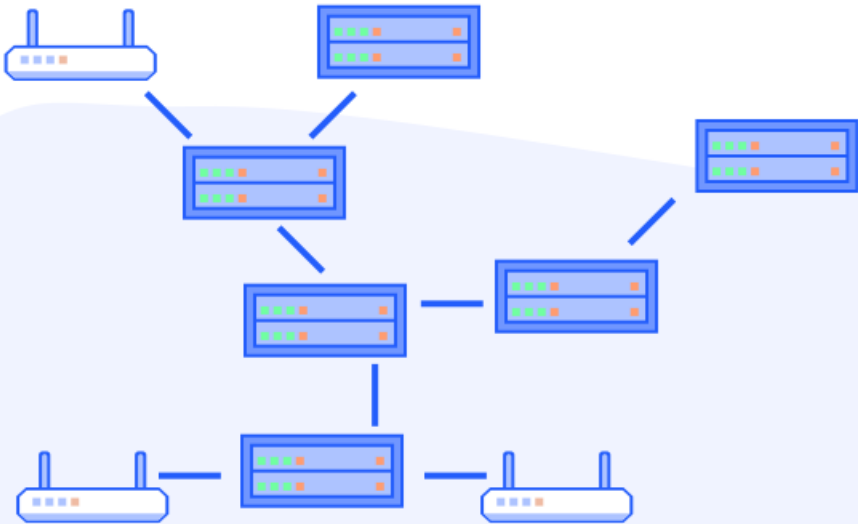
Computers can be connected directly wirelessly or with a cable.



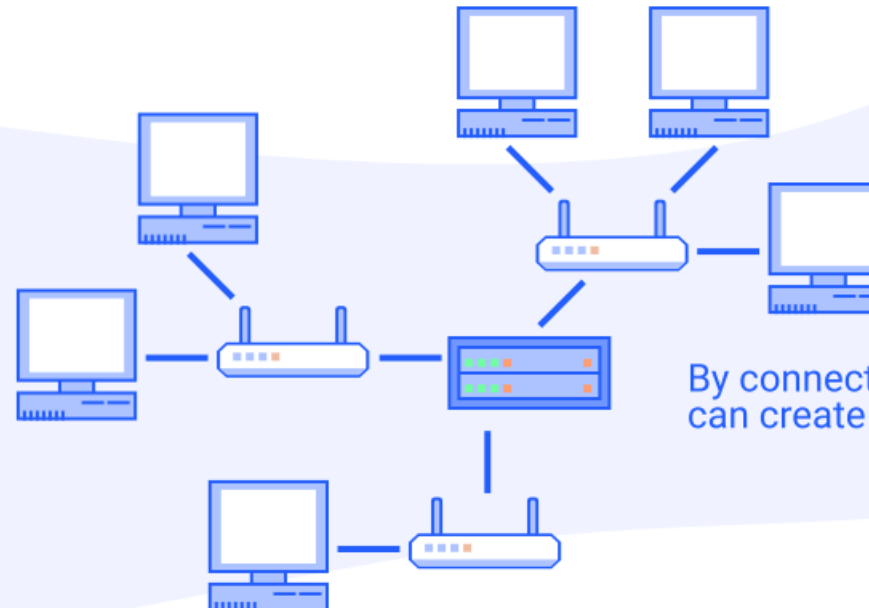
Connecting to every computer in a network becomes complicated very quickly.



Routers are specialized computers that serve as a hub, forwarding information between devices



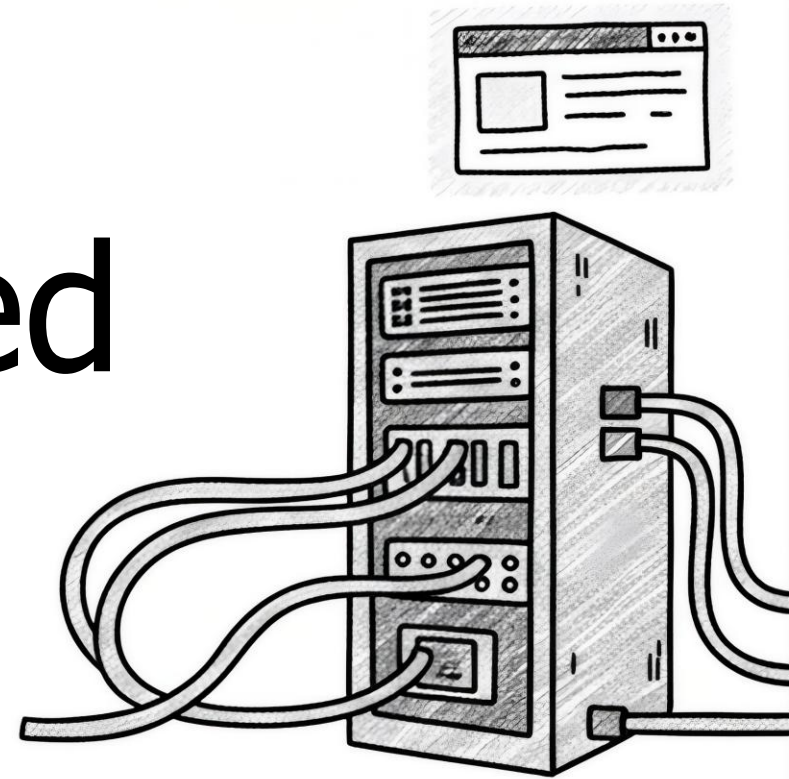
Small networks merged into larger ones, which eventually internetworked to form one worldwide network—the Internet

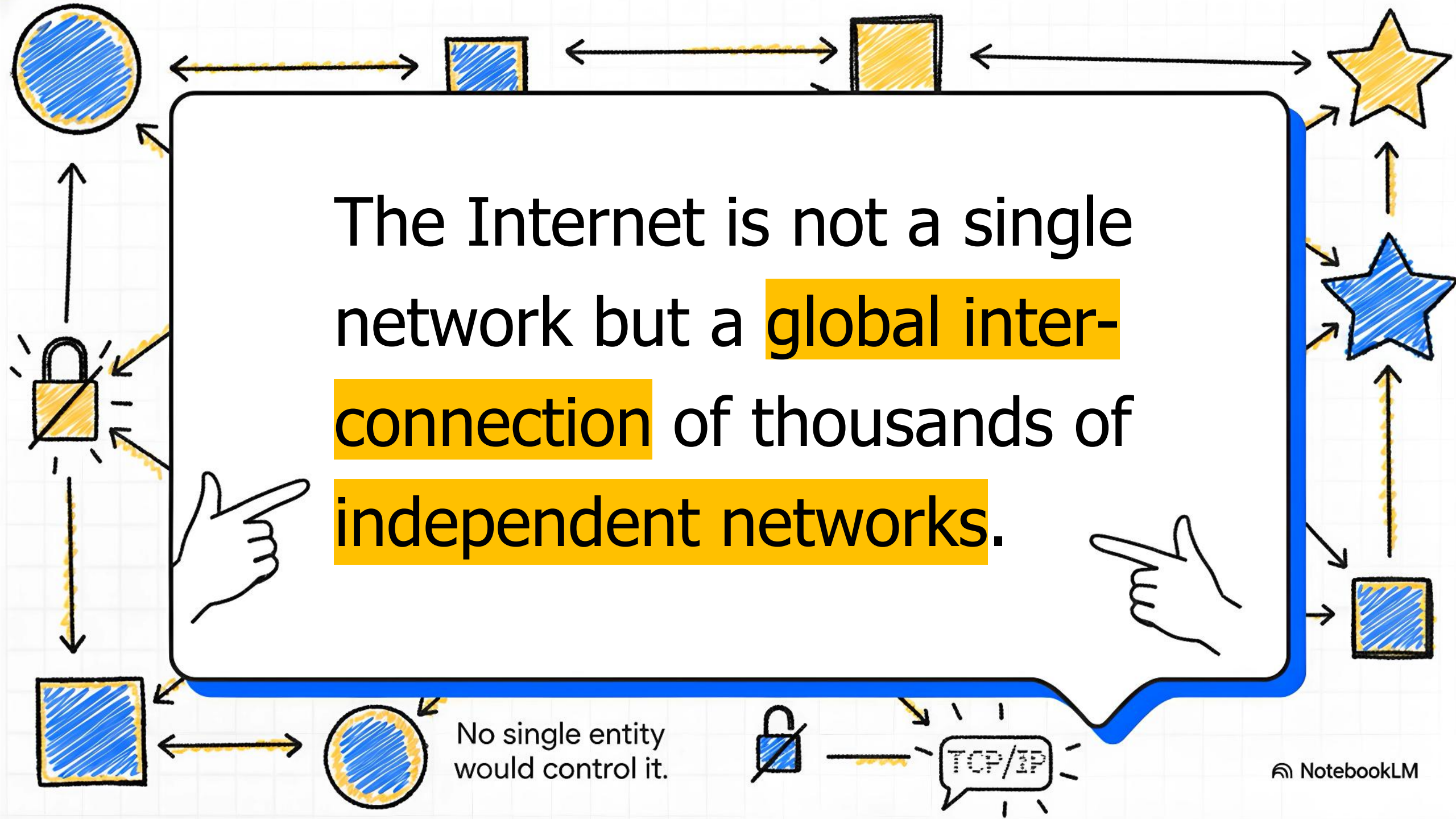


By connecting routers together, we can create a network of networks



Decentralized Miracle





The Internet is not a single network but a **global inter-connection** of thousands of **independent networks**.

No single entity would control it.

TCP/IP

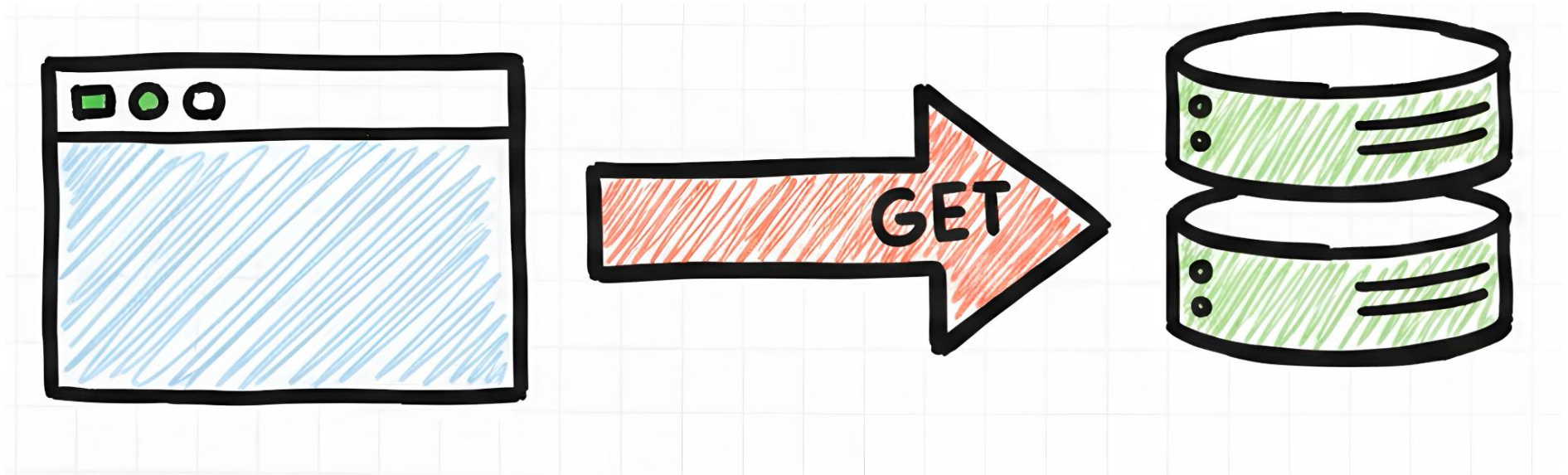
EXPLORING THE HIDDEN INTERNET

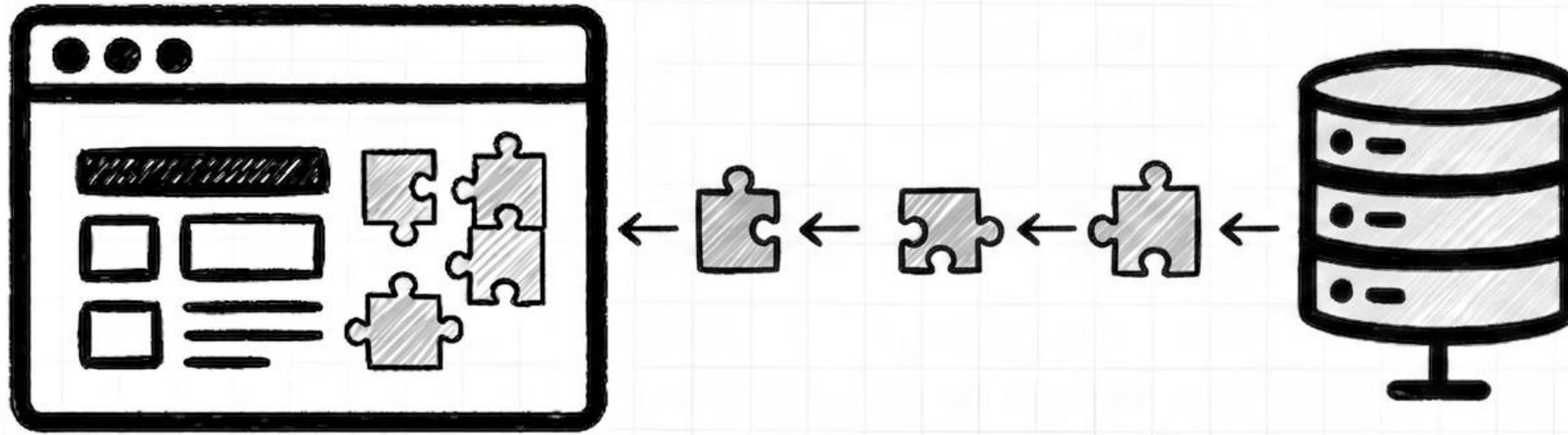




The Destination

Your browser sends an **HTTP** request to the server, using the **GET** command to ask for the contents of the page you requested.





The server responds with the content, consisting of **HTML**, **CSS** and **JavaScript** files. These are assembled into the requested website by the browser.

HTML, CSS and JavaScript

Building blocks of modern websites



HyperText Markup Language (HTML)

[,āch-(,)tē-(,)em-'el]

The basic scripting language used by web browsers to render pages on the world wide web.

What's the Difference?



HTML

Hypertext Markup Language

Create the structure

- Controls the layout of the content
- Provides structure for the web page design
- The fundamental building block of any web page



CSS

Cascading Style Sheet

Stylize the website

- Applies style to the web page elements
- Targets various screen sizes to make web pages responsive
- Primarily handles the "look and feel" of a web page



Javascript

Increase interactivity

- Adds interactivity to a web page
- Handles complex functions and features
- Programmatic code which enhances functionality

HTML Page Structure

`<!DOCTYPE html>` ← Tells version of HTML

`<html>` ← HTML Root Element

`<head>` ← Used to contain page HTML metadata

`<title>Page Title</title>` ← Title of HTML page

`</head>`

`<body>` ← Hold content of HTML

`<h2>Heading Content</h2>` ← HTML heading tag

`<p>Paragraph Content</p>` ← HTML paragraph tag

`</body>`

`</html>`